HERIOT WATT UNIVERSITY

School of Mathematical and Computer Sciences

Actuarial Mathematics and Statistics

F78AB

Actuarial and Financial Mathematics

IN-PERSON EXAMINATION

April Examination and Assessment Diet 2024-2025

Duration: 2 Hours

Total marks available: 60

Answer all SEVEN questions

Resources Included within Exam Paper: No additional attachments
Additional Resources to be Supplied to Candidates: NIL
Calculator Permitted: YES

Ethical Commitment: By submitting answers to this paper, I declare that these answers are entirely my own work and have not been shared, in part or in whole or in any form, with any other student, or disseminated in any way.

I also confirm that I have reviewed and understand all University regulations, policies and procedures on assessment misconduct, including but not limited to accessing unauthorised materials during this examination or the use of unauthorised electronic devices.

I understand that infringing this statement would represent an academic misconduct offence subject to disciplinary action according to the University Regulations and Policy regarding Academic Misconduct, with possible significant consequences for degree progression and final degree outcome.

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1. Determine the present value of an annuity payable where the first payment is £840, due one year from today, and subsequent annual payments decrease by £120 per year. The effective interest rate is 4% compounding annually.

[Total 5 Marks]

2. (a) An investment requires a deposit of £1,000 at time 0, and promises to pay £275 at the end of each year for five years. Using an interest rate of 6% p.a. effective, calculate the discounted payback period for this investment.

[4 marks]

(b) An investment fund was launched on 1 January 2021 with an initial investment of £1,000,000. In subsequent years, an additional £100,000 was added into the fund by investors each 1 January.

The end of year fund balances were as follows:

31 December 2021	£1,200,000
31 December 2022	£1,400,000
31 December 2023	£1,500,000
31 December 2024	£1,800,000

Calculate the time-weighted rate of return earned by this fund during the period 1 January 2021 to 31 December 2024.

[4 marks]

(c) A potential investor has calculated the NPV of a project and found it to be zero. She states *"The project is not worth investing in because it does not make any profit..."*

Comment briefly on her decision, and the rationale she provides for that decision.

[2 marks]

[Total 10 marks]

3. (a) Consider a loan taken for n years at an effective interest rate of i per annum, which is then repaid by level annual instalments of X at the end of each year. Show that the amount of principal repaid in the kth payment can be expressed as $Xv^{n-(k-1)}$.

[3 marks]

(b) You have borrowed £100,000 for 25 years at an effective rate of interest 4% per annum, with a promise to repay level amounts at the end of each year.

i. Show that the level annual repayment is £6,401.20.

[1 mark]

ii. Immediately after the 17th payment, the interest rate is raised to 5% per annum. Calculate the amount of principal repaid in the 20th repayment.

[6 marks]

[Total 10 Marks]

4. The force of interest $\delta(t)$ is a function of time and at any time t (measured in years) is given by:

$$\delta(t) = \begin{cases} 0.035 & 0 < t < 5 \\ 0.02 + 0.0003t^2 & 5 \leq t \leq 15 \end{cases}$$

Calculate the accumulated value at t=15 of a continuous payment stream of £400 per year from time $t = 0$ to $t = 5$.

[Total 6 Marks]

5. Harry purchased a bond with 20 years to redemption. The bond is redeemable at par, and pays coupons of 5% per annum annually in arrears with a gross redemption yield of 6% p.a.. Harry does not pay tax.

(a) Calculate the purchase price per £100 nominal that Harry paid for the bond. [3 marks]

(b) After 10 years, immediately after the payment of the coupon then due, Harry sells the bond to Hermione, who pays income and capital gains tax at a rate of 30%. Hermione purchases the bond to provide a net rate of return of 7% per annum.

i. Calculate the price paid by Hermione. [6 marks]

ii. Calculate the annual effective rate of interest earned by Harry during the period he held the bond. [4 marks]

[Total 13 Marks]

6. In a particular bond market, the two-year par yield at time t=0 is 4.8%, and the issue price at time t=0 of a two year fixed-interest stock, paying coupons of 10% annually in arrears and redeemed at 97 is £106.50 per £100 nominal.

Calculate:

(a) The one-year spot rate

(b) The two-year spot rate

[Total 6 Marks]

7. An insurance company has liabilities of £12M due in 10 years time and £25M due in 15 years time, supported by assets paying £11.657M in 2 years time and £38.813M in 30 years time. The current interest rate is 6% per annum effective.

(a) Show that Redington's first two conditions for immunisation against small changes in the rate of interest are satisfied for this company.

[6 marks]

(b) Determine the profit or loss, expressed as a present value, that the insurance company will make if the interest rates increases immediately to 7% per annum.

[2 marks]

(c) Explain how you might have already anticipated, without making the calculation in part (b), whether the result would be a profit or loss.

[2 marks]

[Total 10 Marks]

END OF PAPER